

F-Test Two-Sample for Variances

Complete line-by-line calculation using Process_A and Process_B

1. Given Data

Observation	Process_A	Process_B
1	52	48
2	55	63
3	49	52
4	58	70
5	61	45
6	54	58
7	57	66
8	53	50
9	59	72
10	50	55
11	56	61
12	60	47
13	51	68
14	55	53
15	58	64

2. Step 1 — Count Observations

Process_A has 15 observations.

Process_B has 15 observations.

Therefore, $df_A = n_A - 1 = 15 - 1 = 14$.

Therefore, $df_B = n_B - 1 = 15 - 1 = 14$.

3. Step 2 — Calculate Mean of Process_A

$Mean_A = \Sigma X_A / n_A = 828 / 15 = 55.2$

So Excel reports Mean = 55.2.

4. Step 3 — Calculate Mean of Process_B

$Mean_B = \Sigma X_B / n_B = 872 / 15 = 58.13333333$

So Excel reports Mean ≈ 58.13333333 .

5. Step 4 — Calculate Variance of Process_A

Sample variance formula: $S^2 = \Sigma(X - \bar{X})^2 / (n - 1)$

X	X – Mean_A	(X – Mean_A) ²
52	-3.2	10.24
55	-0.2	0.04
49	-6.2	38.44
58	2.8	7.84
61	5.8	33.64
54	-1.2	1.44
57	1.8	3.24
53	-2.2	4.84
59	3.8	14.44
50	-5.2	27.04
56	0.8	0.64
60	4.8	23.04
51	-4.2	17.64
55	-0.2	0.04
58	2.8	7.84
Σ		190.40

$$\Sigma(X - \bar{X})^2 = 190.40$$

$$\text{Variance}_A = 190.40 / (15 - 1) = 13.6$$

6. Step 5 — Calculate Variance of Process_B

Sample variance formula: $S^2 = \Sigma(X - \bar{X})^2 / (n - 1)$

X	X – Mean_B	(X – Mean_B) ²
48	-10.133333	102.684444
63	4.866667	23.684444
52	-6.133333	37.617778
70	11.866667	140.817778
45	-13.133333	172.484444
58	-0.133333	0.017778
66	7.866667	61.884444
50	-8.133333	66.151111
72	13.866667	192.284444
55	-3.133333	9.817778
61	2.866667	8.217778

47	-11.133333	123.951111
68	9.866667	97.351111
53	-5.133333	26.351111
64	5.866667	34.417778
Σ		1097.733333

$$\Sigma(X - \bar{X})^2 = 1097.733333$$

$$\text{Variance}_B = 1097.733333 / (15 - 1) = 78.40952381$$

7. Step 6 — Calculate F Statistic

$F = \text{Variance_A} / \text{Variance_B}$

$F = 13.6 / 78.40952381 = \mathbf{0.173448318}$

This matches the Excel output $F \approx 0.173448318$.

8. Step 7 — $P(F \leq f)$ One-Tail

$P(F \leq 0.173448318)$ with $df_{\text{num}} = 14$ and $df_{\text{den}} = 14 = \mathbf{0.001158578}$

This is the left-tail probability reported by Excel as approximately 0.001158578.

9. Step 8 — F Critical One-Tail

At $\alpha = 0.05$, $df_{\text{num}} = 14$ and $df_{\text{den}} = 14$:

$F_{\text{Critical}} = F_{\text{inv}}(0.05; 14, 14) = \mathbf{0.402620943}$

This matches the Excel output F Critical one-tail ≈ 0.402620943 .

10. Step 9 — Decision

$F = 0.173448 < F_{\text{Critical}} = 0.402621$

Because the calculated F is smaller than the one-tail critical value, reject the null hypothesis for this left-tail test.

Conclusion: The two processes do not have equal variances under this one-tailed test. Process_B has a much larger sample variance (≈ 78.41) than Process_A (13.6).

11. Excel Data Analysis Output — Meaning of Each Row

Mean	Average value of each process.
Variance	How spread out the observations are.
Observations	Number of values in each process.
df	Degrees of freedom = $n - 1$.
F	Ratio of the two sample variances.
$P(F \leq f)$ one-tail	Probability in the left tail up to the calculated F.
F Critical one-tail	Critical F value for $\alpha = 0.05$ in the left tail.

Important: Excel's F-Test Two-Sample for Variances uses the first variance divided by the second variance. Here that is $\text{Variance_A} / \text{Variance_B}$, which produces $F \approx 0.173448318$.